**Publicaciones:**

Characterization and sliding wear performance of PMMA reinforced with SiO2 nanoparticles. Journal of Thermoplastic Composite Materials, 2019.

Experimental B-learning laboratory for an electrical machines undergraduate course. IEEE Latin America Transactions, 2016.

Structural analysis of a new total ankle replacement prosthesis with internal structure. Dyna, 2020.

Investigation of the effect of carbonyl iron micro-Particles on the mechanical and rheological properties of isotropic and anisotropic MREs: Constitutive magneto-mechanical material model. Polymers, 2019.

Fabrication and characterization of isotropic and anisotropic magnetorheological elastomers, based on silicone rubber and carbonyl iron microparticles. Polymers, 2018.

Equivalent representation form in the sense of Lyapunov, of nonlinear forced, damped second-order differential equations. Nonlinear Dynamics, 2018.

Evaluation of Surface modification in a Steel track for the rubber tyred Metro. Wear, 2019.

Role of the substrate on the growth of silicon quantum dots embedded in silicon nitride thin films. Materials Chemistry and Physics, 2018.

The SEM electron-mirror effect in human tooth and synthetic hydroxyapatite samples. Microscopy Research and Technique, 2018.

Passive vibration control in a civil structure: Experimental results. Measurement and Control, 2019.

On the boundary conditions in a non-linear dissipative observer for tubular reactors. Processes, 2019.

A nonlinear controller based on saturation functions with variable parameters to stabilize an AUV. International Journal of Naval Architecture and Ocean Engineering, 2019.

Optimum design of a nonlinear vibration absorber coupled to a resonant oscillator: A case study. Shock and Vibration, 2018.

Numerical and Experimental Analyses of the Effect of Heat Treatments on the Phase Stability of Inconel 792. Advances in Materials Science and Engineering, 2018.

Precipitation analysis of as-cast HK40 steel after isothermal aging. International Journal of Minerals, Metallurgy and Materials,2017.

Effect of precipitation on cryogenic toughness in isothermally aged austenitic stainless steel. Metal Science and Heat Treatment, 2017.

**Capítulos de Libro:**

Abrasive Wear Performance of Fe2B Layers Applied on Steel Substrates. Friction, Lubrication and wear, 2019.

## Participación en Congresos:

Active Vibration Suppression in a Flexible Structure by Using a Nonlinear Rotational Actuator. SPIE - The International Society for Optical Engineering, 2020.

SEM and EDS study in bidirectional fabrics on composite materials for aeronautical applications. Microscopy and Microanalysis Annual Conference, 2019.

Nonlinear dynamics of a finite extensibility absorber system examined from its Lyapunov representation form: broaden the frequency bandwidth by exploiting the system internal resonances. First International Nonlinear Dynamics Conference, 2019.

Study of mechanical degradation and microstructural characterization in a Ni-based superalloy component of a gas turbine. THERMEC, 2018.

## Aplicación del Conocimiento:

Patente: Máquina para prueba de fatiga en pavimentos. No: 368677, 2019.

Patente: Máquina de movimiento reciprocante para pruebas de desgaste. No: en trámite, 2020.

Libro Título: Cálculo de Varias Variables. ISBN: 978-607-744-679-8, 2017.

Libro Título: Estática para Ingenieros. ISBN:9786077441366, 2015.